Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/24/2005
PI Name:	Moyers, Michael F. Ph.D.		
Project Title:	HZE Upgrade and Verification of 3-D Transport Code		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	MFMoyers@adelphia.net	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-370-1103
Organization Name:	Loma Linda University Medical Center		
PI Address 1:	Dept. of Radiation Medicine		
PI Address 2:	11234 Anderson Street		
PI Web Page:			
City:	Loma Linda,	**State:**	CA
Zip Code:	92354	**Congressional District:**	41
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	09/12/2003	**End Date:**	08/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Barghouty, N	**Contact Phone:**	256 544-0238
Contact Email:	Nasser.Barghouty@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Waters, Laurie (Los Alamos National Laboratory) James, Michael R. (Los Alamos National Laboratory) McKinney, Greg (Los Alamos National Laboratory) Mashnik, Stephan G. (Los Alamos National Laboratory) Rightnar, Steven (Loma Linda University)		
Grant/Contract No.:	NAG8-1943		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A major threat to long term human space flights such as encountered during extended stay Earth orbiting space station missions or trips to and from Mars is radiation. A large component of this radiation is galactic cosmic rays (CGR). CGR consists of ions as light as hydrogen and as heavy as iron at energies as great as 1 GeV/n. Ions with both high atomic number and high energy are known as HZE particles. The passage of HZE particles through matter is a complex sequence of atomic collisions and nuclear interactions. Optimizing the design of space craft shielding requires simulations of the transport of HZE particles through potential spacecraft configurations and the astronaut's bodies. Monte Carlo (MC) style computer programs are well suited for simulating radiation transport through complex 3-D geometries. One of the most useful MC programs available is MCNPX. Currently, this program is limited to projectile ions with an atomic number less than or equal to 2 whereas HZE consists of ions with atomic numbers greater than 26. The possibility exists to extend MCNPX to these higher atomic number projectiles by implementing a new physics model. A good candidate model for this extension is referred to here as the Los Alamos Quark Gluon String Model (LAQGSM). This work proposes to test the hypothesis that the LAQGSM adequately simulates HZE ion interactions. The specific aims for this project include: 1) modifying the MCNPX code to accept HZE ion projectiles up to 1 GeV/n; 2) modifying the MCNPX code to include the LAQGSM physics ; and 3) testing the ability of the modified code to correctly simulate HZE penetration, scattering, and projectile fragmentation. Specific aim 3 will be performed by comparing model simulations with measure-ments made in an iron ion beam. The experiments and simulations will first be performed in a proton beam having a similar range to validate the test procedures and verify the proton model and data. 1) Provide a user interface to the MCNPX code to input the HZE projectile nucleus type and charge state. 2) Extend calculation model in MCNPX for energy deposition in materials due to ionization to HZE projectiles. 3) Provide a new physics model for nuclear reactions of HZE projectiles with various materials. 4) Provide a nuclear interaction database for use by MCNPX. 5) Provide tallies for both energy deposition and fluxes from the HZE particles. 6) Determine the accuracy of MCNPX for the calculation of ion stopping powers for a medium, energy deposition in the medium, and production of large range secondary particles by measuring depth dose curves in water behind slabs of various materials exposed to both proton and iron beams. 7) Determine the accuracy of MCNPX for calculating the angular scattering power for a medium, the angular distribution of long range secondary particles, and the energy of the long range secondary particles by measuring lateral beam profiles behind slabs of various materials exposed to proton and iron ion beams. see task description
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An accurate three-dimensional particle transport program can also be used to simulate various difficult problems found in particle therapy for cancer. An example includes simulating the perturbations in the dose distribution delievered to a patient with a metal implant treated with a proton or carbon ion beam.
Task Progress:	1) measured depth dose curves in water behind slabs of various materials exposed to proton beam 2) measured depth dose curves in water behind slabs of various materials exposed to iron ion beam 3) measured lateral scatter behind slabs of various materials exposed to proton beam 4) measured lateral scatter behind slabs of various materials exposed to iron ion beam 5) changed user interface to MCNPX Monte Carlo simulation program to input projectile of arbitrary atomic number and mass 6) changed MCNPX Monte Carlo simulation program to calculate energy loss due to ionization from ion projectile of arbitrary atomic number and mass 7) included heavy ion nuclear reaction model calculation routine into MCNPX simulation program, 8) provided tallies for individual ion energy deposition, 9) compared MCNPX calculated dose distributions to measured dose distributions for proton and iron ion beams. In the spring of 2006 the measurements listed above will be repeated for a carbon ion beam.
Bibliography Type:	Description: (Last Updated: 10/24/2005)
Abstracts for Journals and Proceedings	Moyers, M. F.; James, M. F.; Waters, L. S.; McKinney, G. W. "Upgrade and verification of MCNPX for HZE ions" 16th Annual NASA Space Radiation Investigator's Workshop Port Jefferson, New York, May 15-18, 2005 , May-2005
Abstracts for Journals and Proceedings	Moyers, M. F. "Comparison of measured and MCNPX calculated lateral scattering distributions for 250 MeV protons" Space Nuclear Conference San Diego, California, June 4-9, 2005 , Jun-2005